Data Quality and Validation in ETL

1 : Define Data Quality in the context of ETL pipelines. Why is it more than just data cleaning?

- Data Quality in ETL Pipelines refers to the accuracy, completeness, consistency, validity, and reliability of data as it moves through the Extract, Transform, and Load (ETL) process.
- Why is it more than just data cleaning?
- Data cleaning is only one part of data quality. Data quality also involves validating data, checking consistency, removing duplicates, handling missing values, monitoring data over time, and ensuring that data meets business requirements.
- Example: In a sales dataset, data quality ensures that product prices are valid, customer details are complete, sales figures are accurate, and data is consistent across different sources.

2 : Explain why poor data quality leads to misleading dashboards and incorrect decisions.

- Poor data quality leads to misleading dashboards and incorrect decisions because dashboards depend on accurate, complete, and consistent data to display reliable information.

Reasons:

1. Incorrect Data: Wrong values produce inaccurate charts, reports, and business metrics.
2. Missing Data: Incomplete records may lead to incorrect totals, averages, and trends.
3. Duplicate Data: Duplicate records can inflate sales figures, customer counts, or profits.
4. Inconsistent Data: Different formats or units can cause incorrect comparisons between regions or departments.
5. Outdated Data: Old information may lead managers to make decisions based on current conditions incorrectly.

Example: If a sales dashboard contains duplicate transactions, it may show higher sales than the actual amount. A manager might mistakenly increase inventory or set unrealistic sales targets.

3 : What is duplicate data? Explain three causes in ETL pipelines.

- Duplicate Data refers to the presence of the same record more than once in a dataset. It can lead to incorrect calculations, inflated totals, and misleading business reports.
- Three causes of duplicate data in ETL pipelines:
- 1. Repeated Data Extraction: The ETL process may extract the same records multiple times from a source system without checking which records have already been processed.

- 2. Data Integration from Multiple Sources: When combining data from different databases or files, the same customer or transaction may appear in multiple sources.
- 3. Improper Data Loading: If an ETL pipeline runs again after a failure and inserts records without checking existing data, duplicate records may be created.
- Example: If the same sales transaction of ₹5,000 is loaded twice, the dashboard may report ₹10,000 in sales instead of the actual ₹5,000.

4 : Differentiate between exact, partial, and fuzzy duplicates.

- Difference between Exact, Partial, and Fuzzy Duplicates

Type	Meaning	Example
Exact Duplicates	Records that are identical in every field.	`101, Rahul, 5000` appears twice.
Partial Duplicates	Records that match in some fields but differ in others.	`101, Rahul, 5000` and `101, Rahul, 6000`.
Fuzzy Duplicates	Records that are almost identical but contain small spelling or formatting differences.	`Rahul Sharma` and `Rahul Shama`.

Explanation:

1. Exact duplicates: These can often be detected by comparing all columns.
2. Partial duplicates: These can be identified by comparing selected fields, such as customer ID or email address.
3. Fuzzy duplicates: These require approximate matching techniques, such as string similarity, to identify records that likely refer to the same entity.

5 : Why should data validation be performed during transformation rather than after loading?

Data Quality and Validation in ETL

- Data validation should be performed during the transformation stage of an ETL pipeline to identify and correct errors before the data is loaded into the target system.

Reasons:

1. Prevents Incorrect Data: Invalid, incomplete, or inconsistent data can be detected before entering the target database.
2. Maintains Data Quality: Ensures that data follows the required formats, rules, and business conditions.
3. Reduces Errors and Costs: Fixing problems during transformation is generally easier and less expensive than correcting them after loading.
4. Protects the Target System: Prevents unreliable data from affecting dashboards, reports, and business decisions.
5. Improves Efficiency: Invalid records can be corrected, rejected, or sent to an error table before loading.

Example: If an employee's salary is recorded as `-5000`, the ETL pipeline can validate the salary during transformation and reject or flag the invalid record before loading it into the employee database.

6 : Explain how business rules help in validating data accuracy. Give an example.

- Business rules are predefined conditions or guidelines used to ensure that data is accurate, valid, and consistent with an organization's requirements during an ETL pipeline.

How business rules help in data validation:

1. Ensure Accuracy: Check whether values are reasonable and correct according to business requirements.
2. Maintain Consistency: Ensure that data follows the same standards across different systems.
3. Detect Errors: Identify invalid values, missing information, and unusual records.
4. Support Decision-Making: Ensure that reports and dashboards use reliable data.

Example: In an employee database, a business rule states that an employee's salary must be greater than zero. If a record contains a salary of ₹−5,000, the ETL pipeline flags it as invalid and prevents it from being loaded until it is corrected.